Shorter question part

(1) (a) Find $A \cap B \cap C$ if $A = \{x : 0 \leq x \leq 6\}$, $B = \{x : 4 \leq x\}$ and $C = \{x = 0, 1, 2, 3, \ldots\}$.

Solution: $A \cap B = \{x : 4 \leq x \leq 6\}$, taking intersection with C, $A \cap B \cap C$={4,5,6}.

(b) Ten basketball players meet in the school gym for a pickup game. How many ways can they form a blue team and a red team of five players each?

Solution: This is equivalent to choosing 5 players from 10. $\binom{10}{5} = 252$.

(2) (a) A desk has three drawers. The first contains two gold coins, the second has two silver coins, and the third has one gold coin and one silver coin. A coin is drawn at random from a drawer selected at random. Suppose the coin selected was silver. What is the probability that the other coin in that drawer is gold?

Solution: Let A_i be the event that drawer i is chosen, $i = 1, 2, 3$. If B is the event a silver coin is selected, then

$$P(A_3|B) = \frac{0.5(1/3)}{0(1/3) + 1(1/3) + 0.5(1/3)} = 1/3$$

(b) A bleary-eyed student awakens one morning, late for an 8:00 class, and pulls two socks out of a drawer that contains two black, four brown, and two blue socks, all randomly arranged. What is the probability that the two he draws are a matched pair?

Solution: Total number of pairs is $\binom{8}{2} = 28$, there are $\binom{2}{2} + \binom{4}{2} + \binom{2}{2} = 8$ matched pairs, thus the probability is $8/28 = 0.286$.

(3) Let X have a Poisson distribution with a mean of 5. Find $P(2 \leq X < 4)$.

Solution:

Use $P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$ with $\lambda = 5$ and $x = 2, 3$. This gives

$$P(2 \leq X < 4) = P(X = 2) + P(X = 3) = \frac{e^{-5}5^2}{2!} + \frac{e^{-5}5^3}{3!} = 0.2246$$

(4) It is known that the probability is 0.3 that a child exposed to a certain contagious disease will catch it. What is the probabality that the 10th child exposed to the disease will be the fourth to catch?

Solution:

Use If $X \sim NB(x, k, p) = \binom{x-1}{k-1} p^k (1-p)^{x-k}$ with $x = 10,\ k = 4$ and $p = 0.3$.
Required probability= $\binom{9}{3}(0.3)^4(0.7)^6 = 0.0800$

(5) Let $Y_1, Y_2, \cdots, Y_9$ be a random sample of size 9 from a continuous distribution with pdf $f_Y(y) = 2y,\ 0 \leq y \leq 1$. Given that its cfd is $F_Y(y) = y^2,\ 0 \leq y \leq 1$, find the pdf of Y_3', the third smallest in the sample.

Solution:
Substitute $n = 9$ and $i = 3$ with given pdf and cdf into

$$f_{Y_i'}(y) = \frac{n!}{(i-1)!(n-i)!}[F_Y(y)]^{i-1}[1 - F_Y(y)]^{n-i} f_Y(y)$$

and get

$$f_{Y_3'} = \frac{9!}{2!6!}[y^2]^2[1-y^2]^6(2y) = 504y^5(1-y^2)^6.$$

(6) Let $f_{XY}(x,y)$ be the joint pdf of two continuos random variables given by $f_{XY}(x,y) = 2,\ x+y<1, x\geq 0; y>0$. Find $f_{Y|x}(y|x)$, the conditional density of Y given $X=x$.

Solution.
First obtain $f_X(x)$, the marginal density of X. That is,

$$f_X(x) = \int_0^{1-x} 2dy = 2(1-x),\ 0\leq x<1.$$

Therefore, $f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{1}{1-x};\ x+y<1; 0\leq x<1; y>0.$

Longer question part

(7) A total of n different letters, say addressed to person $P_1, P_2, \ldots, P_n$ are assigned at random to n labeled envelopes, say with address $A_1, A_2, \ldots, A_n$.

(a) If $n = 3$, characterise all sample outcomes in the sample space S_3. (Hint: write (2,1,3) for the sample outcome that A_1 contains letter to P_2, A_2 contains letter to P_1 and A_3 letter to P_3.)

Solution. $S_3 = \{(1,2,3),(1,3,2),(2,1,3),(2,3,1),(3,1,2),(3,2,1)\}$.

(b) For any $n \geq 2$, show that the number of sample outcomes in the sample space is $S_n = n!$.

Solution. Using the multiplication rule, there are n possibilities for the first number, $(n-1)$ for the second number etc, thus there are $n!$ possible sample outcomes in S_n.

(c) Let the random variable $X_i \in \{0,1\}$ and $X_i = 1$ when envelope with address A_i contains letter to P_i and $X_i = 0$ otherwise. Show that

$$p_{X_i}(k) = P(X_i = k) = \begin{cases} \frac{n-1}{n} & \text{for } k = 0 \\ \frac{1}{n} & \text{for } k = 1 \end{cases}.$$

Solution. Without loss of generality let $i = n$. There are $n-1$ possible P_j's for envelope A_1, $n-2$ for A_2 etc, thus $(n-1)!$ favourable outcomes among the $n!$ possible outcomes. Using the classical definition of probability $P(X_i = 1) = \frac{(n-1)!}{n!} = 1/n$. $P(X_i = 0) = 1 - P(X_i = 1) = \frac{n-1}{n}$.

(d) Derive the joint distribution of (X_i, X_j) for $i \neq j$. (Hint: $P(X_i = 1, X_j = 1) = \frac{1}{(n-1)n}$.)

Solution. Similar to before, without loss of generality, let $i = n-1$ and $j = n$, then

$$p_{X_i,X_j}(k_i,k_j) = P(X_i = k_i, X_j = k_j) = \begin{cases} \frac{(n-2)!}{n!} = \frac{1}{(n-1)n} & \text{for } k_i = 1,\ k_j = 1 \\ \frac{(n-2)(n-2)!}{n!} = \frac{(n-2)}{(n-1)n} & \text{for } k_i = 1,\ k_j = 0 \\ \frac{(n-2)(n-2)!}{n!} = \frac{(n-2)}{(n-1)n} & \text{for } k_i = 0,\ k_j = 1 \\ \frac{n(n-1)-2(n-2)-1}{(n-1)n} = \frac{n^2-3n+3}{(n-1)n} & \text{for } k_i = 0,\ k_j = 0 \end{cases}$$

(e) Let $X = X_1 + X_2 + \ldots + X_n$, calculate (i) $\mathrm{E}(X)$, (ii) $\mathrm{E}(X_1X_2)$ and (iii) $\mathrm{Var}(X_1 + X_2)$.

Solution. (i) We have that $E(X) = E(X_1 + X_2 + ... + X_n) = E(X_1) + E(X_2) + ... + E(X_n) = nE(X_1) = n\frac{1}{n}$, because $E(X_1) = 1p_{X_1}(1) = 1/n$.
(ii)$E(X_1X_2) = (1)(1)P(X_1 = 1, X_2 = 1) = \frac{1}{n(n-1)}$.
(iii) $\mathrm{Var}(X_1 + X_2) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2) + 2\,\mathrm{Cov}(X_1, X_2) = 2E(X_1^2) - 2[E(X_1)]^2 + 2E(X_1X_2) - 2E(X_1)E(X_2)$.. $E(X_1^2) = 1/n$, we have

$$\mathrm{Var}(X_1 + X_2) = \frac{2}{n} - \frac{2}{n^2} + \frac{2}{n(n-1)} - \frac{2}{n^2} = \frac{2(n^2 - 2(n-1))}{n^2(n-1)}.$$

(8) (D) Let $\overline{X}_1$ and $\overline{X}_2$ be the means of two independent samples of sizes n_1 and n_2 from a normal population with the mean μ and variance σ^2. Show that $U = \omega\overline{X}_1 + (1-\omega)\overline{X}_2$ is also unbiased for μ, where $0 < \omega < 1$ is a constant. Find the value of ω such that Var(U) is minimum.

Solution:

$$E[U] = \omega E[\overline{X}_1] + (1-\omega)E[\overline{X}_2] = \omega\mu + (1-\omega)\mu = \mu.$$

$Var(U) = \omega^2 Var[\overline{X}_1] + (1-\omega)^2 Var[\overline{X}_2] = \omega^2\frac{\sigma^2}{n_1} + (1-\omega^2)^2\frac{\sigma^2}{n_2}$.

$$\frac{\partial Var(U)}{\partial \omega} = 0$$

gives $\omega = \frac{n_1}{n_1+n_2}$. Now

$$\frac{\partial^2 Var(U)}{\partial \omega^2} = \frac{1}{n_1} + \frac{1}{n_2}$$

is always positive. Hence at $\omega = \frac{n_1}{n_1+n_2}$, we have minimum $Var(U)$.

(1) Consider two continuous random variables X and Y with joint pdf

$$f_{XY}(x,y) = \begin{cases} \frac{2}{81}x^2y & 0 < x < K,\ 0 < y < K \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the value of K so that $f_{XY}(x,y)$ is a valid joint pdf.
Solution: We have that

$$1 = \int_0^K \int_0^K \frac{2}{81}x^2 y dx dy = \frac{k^5}{243} \to K = 3.$$

(b) Find $P(X > 3Y)$.
Solution:

$$P(X > 3Y) = \int_0^3 \int_0^{x/3} \frac{2}{81}x^2 y dy dx = \int_0^3 \frac{1}{729}x^4 dx = \frac{1}{15}.$$

(c) Find $P(X + Y > 3)$.
Solution:

$$\begin{aligned} P(X+Y>3) &= \int_0^3 \int_{3-x}^3 \frac{2}{81}x^2 y dy dx \\ &= \int_0^3 \frac{1}{81}x^2[9-(3-x^2)]dx \\ &= \frac{1}{81}\int_0^3 (6x^3 - x^4)dx \\ &= \frac{1}{81}\left(\frac{3}{2}x^4 - \frac{1}{5}x^5\right)\Bigg|_0^3 \\ &= \frac{1}{81}\left(\frac{243}{2} - \frac{243}{5}\right) = 0.90 \end{aligned}$$

(d) Are X and Y independent? Find $\mathrm{Cov}(X, Y)$.
Solution:

$$
\begin{aligned}
f_X(x) &= \int_0^3 \frac{2}{81} x^2 y dy = \frac{1}{9} x^2 \qquad 0 < x < 3 \\
f_Y(y) &= \int_0^3 \frac{2}{81} x^2 y dx = \frac{2}{9} y \qquad 0 < y < 3 \\
f_{XY}(x, y) &= f_X(x) \cdot f_Y(y) = \frac{1}{9} x^2 \frac{2}{9} y
\end{aligned}
$$

then we can say that X and Y are independent and, therefore, $Cov(X, Y) = 0$.